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SpaceCDF — Installation Guide

From GitHub to your first design in 30 minutes

SpaceCDF

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Faculty of Engineering • School of Engineering Design and Teaching Innovation (SEDTI)

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CPU	Any 64-bit dual-core
RAM	4 GB
Disk space (just the tool)	700 MB
Disk space (with full ECSS standards library)	1.5 GB
Python	3.11 or 3.12
Node.js	18 LTS
Browser	Chrome 120, Firefox 120, Safari 17, Edge 120

1.2 Where the disk space goes

These are the actual measured footprints after a clean install:

Component	Size	Optional?
Source code (`packages/`, `frontend/`, `docs/`, `scripts/`)	≈ 200 MB	Required
Python virtual environment (backend dependencies)	≈ 230 MB	Required
Frontend `node_modules` (build tools)	≈ 170 MB	Required
ECSS active-standards PDFs	≈ 145 MB	Optional
ECSS handbooks archive	≈ 370 MB	Optional

You can safely delete the two ECSS PDF folders after installation if you do not need the standards offline.

1.3 Performance

On a typical laptop the design loop converges in **under 100 milliseconds** for a small mission and the full document bundle (Word + Excel + Markdown with embedded figures) renders in **about one second**. Peak RAM use is ≈ **150 MB** per active mission.

2. Install Prerequisites

You need three pieces of supporting software before you can install SpaceCDF itself.

2.1 Install Python 3.11+

macOS

1. Open the Terminal application (press ⌘ -Space, type **Terminal**, press Return).

2. Type the following and press Return:

```
python3 --version
```

3. If the response is `Python 3.11.x` or higher, you are done with this step. Otherwise, continue.

4. Install ***Homebrew*** if you do not already have it. Paste this single line into the Terminal and press Return:

```
/bin/bash -c "$(curl -fsSL  
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

5. Install Python via Homebrew:

```
brew install python@3.12
```

Windows 11

6. Open your browser and go to <<https://www.python.org/downloads/windows/>>.

7. Click ***Download Python 3.12*** (or 3.11 — either works).

8. ***Important.*** When the installer opens, ***tick the box that says*

*"Add Python to PATH"**** before clicking *Install Now*.*

9. After installation, open ***PowerShell*** (press the Windows key, type **powershell**, press Return) and type:

```
python --version
```

You should see Python 3.12.x.

Ubuntu Linux 22.04+

```
sudo apt update  
sudo apt install -y python3 python3-pip python3-venv  
python3 --version
```

2.2 Install Node.js 18+

Node.js builds the SpaceCDF web interface.

macOS

```
brew install node  
node --version
```

Windows 11

10. Go to <<https://nodejs.org/en/download>> and download the ****LTS**** Windows installer.

11. Run the installer and accept all defaults.

12. Open a ****new**** PowerShell window (any window opened before installation will not see the new Node.js) and type:

```
node --version
```

You should see v20.x.x or v22.x.x.

Ubuntu Linux

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt install -y nodejs  
node --version
```

2.3 Install Git

Git is the program SpaceCDF uses to download itself from GitHub.

macOS

Git ships with the developer command-line tools. Type in Terminal:

```
git --version
```

If it asks you to install the developer tools, click **Install** and wait about five minutes.

Windows 11

13. Go to <<https://git-scm.com/download/win>>.

14. Run the installer and accept all defaults.

15. After installation, open a new PowerShell and type:

```
git --version
```

Ubuntu Linux

```
sudo apt install -y git
```

```
git --version
```

*****Stuck?**** At this point you should have three working commands:*

```
`python3 --version` (3.11 or later), `node --version` (18 or later),
```

and `git --version`. If any one of these still fails after restarting

*the terminal, jump to §7 ***Troubleshooting*** before continuing.*

3. Download SpaceCDF

Choose a folder where you keep your projects. In the examples below the folder is your home directory (~); change it to wherever you like.

macOS / Linux

```
cd ~  
git clone https://github.com/FNewland/SpaceCDF.git  
cd SpaceCDF
```

Windows 11

```
cd $HOME  
git clone https://github.com/FNewland/SpaceCDF.git  
cd SpaceCDF
```

This downloads about 200 MB of source code and may take a few minutes on a slow connection.

4. Install the Backend (Python)

The backend is the engine that runs the 20 design agents and produces documents.

4.1 Create a Python virtual environment

A **virtual environment** keeps SpaceCDF's Python dependencies isolated from

anything else on your computer.

macOS / Linux

```
cd ~/SpaceCDF
python3 -m venv .venv
source .venv/bin/activate
```

Windows 11

```
cd $HOME\SpaceCDF
python -m venv .venv
.venv\Scripts\Activate.ps1
```

If PowerShell refuses with an "execution policy" error, run this **one time only** in an **Administrator** PowerShell window:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Then close and reopen PowerShell and try the activation again.

Verify. Once activated, the prompt should be prefixed with (.venv).

4.2 Install the SpaceCDF Python packages

With the virtual environment active, install the four backend packages.

This downloads about 230 MB of dependencies and takes about 5 minutes.

```
pip install --upgrade pip
pip install -e packages/spacecdf-common
pip install -e packages/spacecdf-agents
pip install -e packages/spacecdf-kb
pip install -e packages/spacecdf-server
```

You should see "Successfully installed ..." four times.

4.3 Smoke-test the backend

```
python scripts/run_design.py configs/examples/6u_eo_cubesat.yaml
```

After about a second you should see a `MISSION: EOSAT-1 Multispectral Imager` banner and a printout of mass, power, cost, and other budgets, followed by CONVERGED. If you reach this point the backend is working.

5. Install the Frontend (Web Interface)

In a **second** terminal window, leave the first one running, navigate to the frontend folder, and install the JavaScript build tools.

```
cd ~/SpaceCDF/frontend      # or: cd $HOME\SpaceCDF\frontend
npm install
```

npm install downloads about 170 MB and takes 3–6 minutes the first time. You may see warnings — these are normal as long as the command finishes with a added <N> packages line.

6. First Launch

You need **two terminals open at once**: one for the backend server and one for the frontend.

6.1 Start the backend (Terminal 1)

```
cd ~/SpaceCDF      # or: cd $HOME\SpaceCDF
source .venv/bin/activate  # macOS / Linux
# .venv\Scripts\Activate.ps1 # Windows
uvicorn spacecdf_server.app:app --reload --port 8000
```

You should see lines ending in Uvicorn running on `http://0.0.0.0:8000`. Leave this terminal open.

6.2 Start the frontend (Terminal 2)

```
cd ~/SpaceCDF/frontend      # or: cd $HOME\SpaceCDF\frontend
npm run dev
```

You should see a line ending in Local: `http://localhost:5173/`.

6.3 Open SpaceCDF in your browser

Open `<http://localhost:5173>` in Chrome, Firefox, Safari, or Edge. You should see the SpaceCDF welcome screen.

*****Success criterion.** On the welcome screen, click **New study**. If***

you can name a study and reach Step 1 (*Mission Need*) without an error

banner, SpaceCDF is fully installed.

6.4 Run your first design

16. From the welcome screen click ****New study****. Name it `My First Mission`.
 17. In Step 1 (***Mission Need***), pick any sample problem from the suggestions list, then click **Continue**.
 18. Step 2 (***Concept***): accept the default.
 19. Step 3 (***Requirements***): pick "SSO at 500 km" and one payload.
 20. Step 4 (***Design***): click the big ****Run Design**** button. The 20 agents converge in under a second.
 21. Open the ****Exports**** tab and click ****Generate SRR****. After a second a .docx, .xlsx, and .md bundle is offered as a download.
- If you reach this point, congratulations — you are running a complete concurrent design facility on your own machine.

7. Troubleshooting

"Command not found: python3" (macOS / Linux)

You skipped §2.1. Run that section again, then close and reopen the terminal so it picks up the new PATH.

"Activate.ps1 cannot be loaded because running scripts is disabled" (Windows)

You need to allow PowerShell scripts. In an Administrator PowerShell:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Then reopen PowerShell and try the activation again.

`pip install` errors mentioning `wheel` or `setuptools`

Update pip first, then retry:

```
pip install --upgrade pip setuptools wheel
```

`npm install` errors

If the install fails partway through, delete the partial install and try again:

```
rm -rf node_modules package-lock.json # macOS / Linux
# Remove-Item -Recurse -Force node_modules,package-lock.json # Windows
npm install
```

"Port 8000 already in use" or "Port 5173 already in use"

Another program is using that port. Either close it or pick a different port:

```
uvicorn spacecdf_server.app:app --reload --port 8100
```

If you change the backend port, also update frontend/vite.config.ts so the frontend knows where to find it.

The browser shows "ERR_CONNECTION_REFUSED"

Either the backend (port 8000) or the frontend (port 5173) is not running. Switch to each terminal and confirm both are still alive.

Designs do not converge

Look at the warnings panel in Step 4. A common cause is asking for a payload that draws more power than the spacecraft class can carry — pick a smaller payload or a larger spacecraft class.

Document export fails

Make sure your virtual environment is the one in which you installed the packages. The prompt should say (.venv). If not, re-activate it.

8. Optional Add-ons

8.1 Run with PostgreSQL instead of in-memory storage

By default SpaceCDF keeps studies in memory only — they vanish when you restart the server. To persist them, install PostgreSQL 15+ and add a database URL to your environment:

```
export SPACECDF_DATABASE_URL=postgresql://localhost/spacecdf
```

PostgreSQL adds about 200 MB to disk and ~100 MB to RAM.

8.2 Remote access via Tailscale

If you want to share your local instance with colleagues, install Tailscale (<<https://tailscale.com/download>>) on both machines and use your tailnet IP. Detailed steps are in docs/REMOTE_ACCESS.md.

8.3 Keep the ECSS standards library offline

The two ECSS folders (Active ECSS Standards_PDF-files ... and ECSS-Handbooks_...) take about 500 MB. If you want them, keep them in the repository root. If not, you can move them to an external drive or delete them — the tool falls back to citing the standards by reference.

9. Updating SpaceCDF

To pull the latest version from GitHub:

```
cd ~/SpaceCDF          # or: cd $HOME\SpaceCDF
git pull
source .venv/bin/activate # macOS / Linux
pip install -e packages/spacecdf-common -e packages/spacecdf-agents -e packages/spacecdf-kb -e
packages/spacecdf-server
cd frontend
npm install
```

Then restart the two terminals from §6.

10. Uninstalling

SpaceCDF lives entirely inside its own folder and the Python virtual environment. To remove it completely:

```
# macOS / Linux
rm -rf ~/SpaceCDF
```

```
# Windows
Remove-Item -Recurse -Force $HOME\SpaceCDF
```

The Python, Node.js, and Git installations are independent and remain on your machine for other software to use.

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